DAA (LAB)

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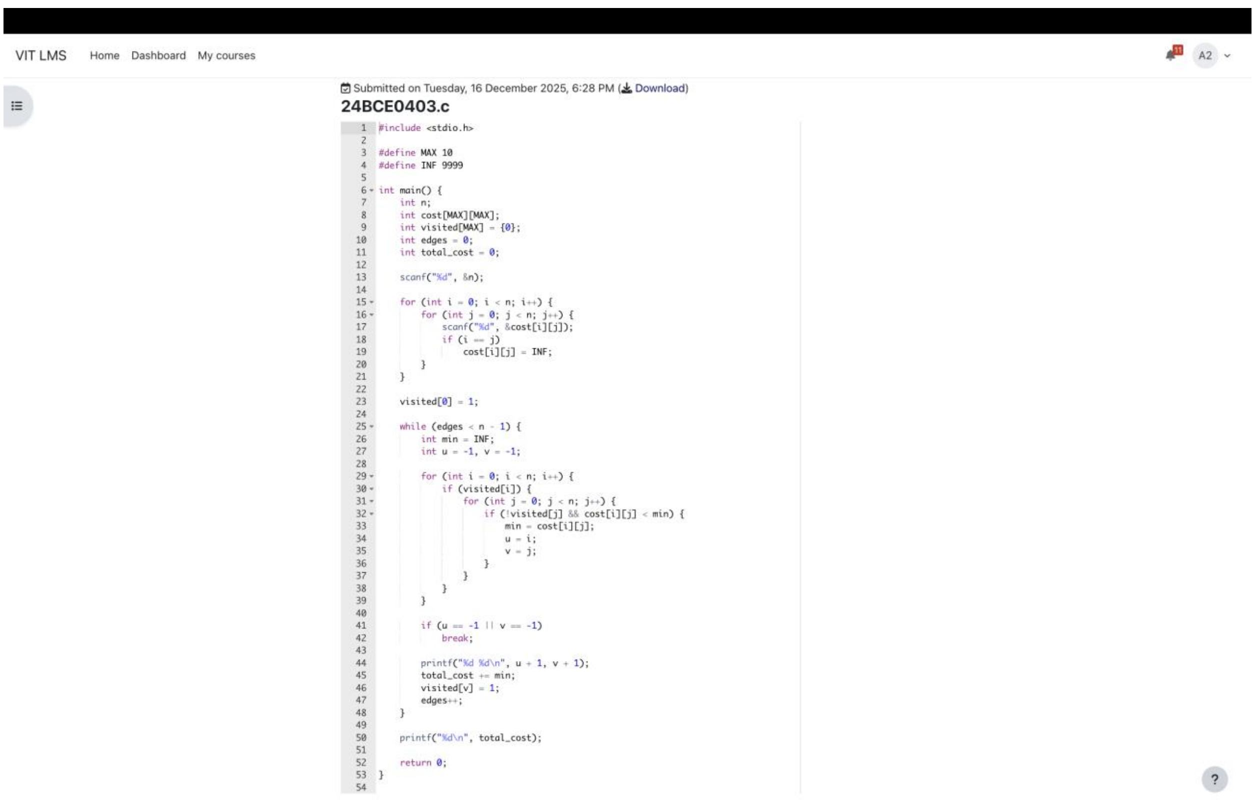
EXPERIMENt:- DA1

DATE:- 4-MAR-2026

PRIMS ALGORITHM:-



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Submitted on Tuesday, 16 December 2025, 6:28 PM (Download)

24BCE0403.c

```c
#include <stdio.h>

#define MAX 10
#define INF 9999

int main() {
    int n;
    int cost[MAX][MAX];
    int visited[MAX] = {0};
    int edges = 0;
    int total_cost = 0;

    scanf("%d", &n);

    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            scanf("%d", &cost[i][j]);
            if (i == j)
                cost[i][j] = INF;
        }
    }

    visited[0] = 1;

    while (edges < n - 1) {
        int min = INF;
        int u = -1, v = -1;

        for (int i = 0; i < n; i++) {
            if (visited[i]) {
                for (int j = 0; j < n; j++) {
                    if (!visited[j] && cost[i][j] < min) {
                        min = cost[i][j];
                        u = i;
                        v = j;
                    }
                }
            }
        }

        if (u == -1 || v == -1)
            break;

        printf("%d %d\n", u + 1, v + 1);
        total_cost += min;
        visited[v] = 1;
        edges++;
    }

    printf("%d\n", total_cost);

    return 0;
}
```

KRUSKALS ALGORITHM:-

[redacted]

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A2

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24BCE0403.c

```c
#include <stdio.h>
#include <stdlib.h>

struct Edge {
    int u, v, w;
};
int parent[100];

int find(int i) {
    if (parent[i] != i)
        parent[i] = find(parent[i]);
    return parent[i];
}

void unite(int a, int b) {
    int rootA = find(a);
    int rootB = find(b);
    parent[rootA] = rootB;
}

int compare(const void *a, const void *b) {
    return ((struct Edge *)a)->w - ((struct Edge *)b)->w;
}

int main() {
    int n, e;
    struct Edge edges[100];
    scanf("%d", &n);
    scanf("%d", &e);

    for (int i = 0; i < e; i++) {
        scanf("%d %d %d", &edges[i].u, &edges[i].v, &edges[i].w);
    }


    for (int i = 1; i <= n; i++)
        parent[i] = i;


    qsort(edges, e, sizeof(struct Edge), compare);

    int mstCost = 0, count = 0;

    for (int i = 0; i < e && count < n - 1; i++) {
        int u = edges[i].u;
        int v = edges[i].v;

        if (find(u) != find(v)) {
            unite(u, v);
            printf("%d %d %d\n", u, v, edges[i].w);
            mstCost += edges[i].w;
            count++;
        }
    }

    printf("\nTotal MST cost %d\n", mstCost);
```

FRACTIONAL KNAPSACK PROBLEM:-

[redacted]

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A2

Reviewed on Tuesday, 6 January 2026, 10:49 AM by Automatic grade
Grade: 100.00 / 100.00

Assessment report [-]
[+]**Summary of tests**

Submitted on Tuesday, 6 January 2026, 10:48 AM (Download)

aryan.cpp

```cpp
#include <iostream>
#include <vector>
#include <algorithm>
#include <iomanip>

using namespace std;

struct vegetable{
    int id;
    double weight;
    double profit;
    double ratio;
};
bool compare(vegetable a, vegetable b){
    return a.ratio>b.ratio;

}
int main(){
    int n;
    if(!(cin>>n))return 0;
    vector<vegetable> veggies(n);
    for (int i=0;i<n;i++){
        cin>> veggies[i].weight;
        veggies[i].id=i;
    }
    for(int i=0;i<n;i++){
        cin>>veggies[i].profit;
        veggies[i].ratio=veggies[i].profit/veggies[i].weight;
    }
    double capacity;
    cin>>capacity;
    sort(veggies.begin(),veggies.end(),compare);
    double totalProfit=0.0;
    for(int i=0;i<n;i++){
        if(capacity>=veggies[i].weight){
            capacity-=veggies[i].weight;
            totalProfit+=veggies[i].profit;
        }
        else{
            double fraction = capacity/veggies[i].weight;
            totalProfit += veggies[i].profit*fraction;
            break;
        }
    }
    cout << fixed << setprecision(2) << totalProfit<<endl;
    return 0;

```

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HUFFMAN CODING:-

[redacted]

Submitted on Wednesday, 7 January 2026, 12:22 AM (Download) (Evaluate)

Automatic evaluation [-]

Proposed grade: 50 / 100

Comments [-]

[+]Test 1: huffman

[+]Summary of tests

huffman.cpp

```cpp
#include <iostream>
#include <vector>
#include <queue>
#include <string>

using namespace std;


struct Node {
    char ch;
    int freq;
    long long id;
    Node *left, *right;

    Node(char character, int frequency, long long creation_id) {
        ch = character;
        freq = frequency;
        id = creation_id;
        left = right = nullptr;
    }
};


struct compare {
    bool operator()(Node* l, Node* r) {
        if (l->freq != r->freq) {
            return l->freq > r->freq;
        }
        return l->id > r->id;
    }
};

void generateCodes(Node* root, string str) {
    if (!root) return;
    if (!root->left && !root->right) {
        cout << root->ch << " " << str << endl;
        return;
    }
    generateCodes(root->left, str + "0");
    generateCodes(root->right, str + "1");
}

int main() {
    int n;
    if (!(cin >> n)) return 0;

```

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